

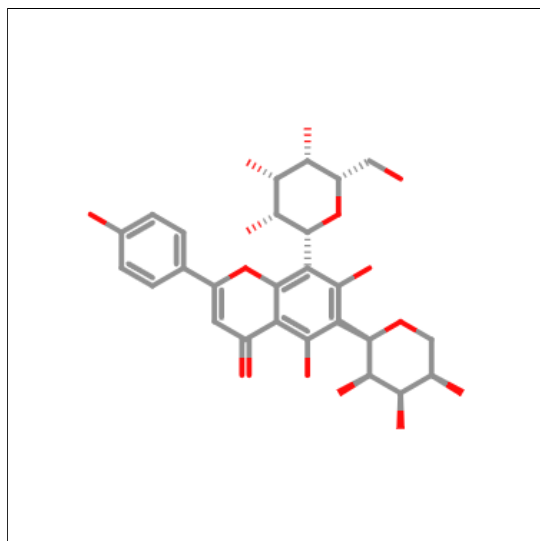
Toxicity Model Computation

If you want to close this window, you can get your results later with this link :

[Access results](#) (please copy link location)

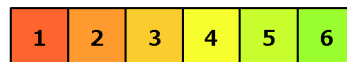
Model computation is complete. You can view the toxicity radar chart here (opens a new tab) [Toxicity Radar Chart](#) and the network chart here [Network Chart](#)

Oral toxicity prediction results for input compound



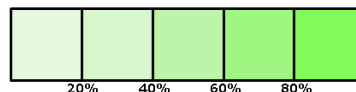
Predicted LD50: 832mg/kg

Predicted Toxicity Class: 4



Average similarity: 57.81%

Prediction accuracy: 67.38%



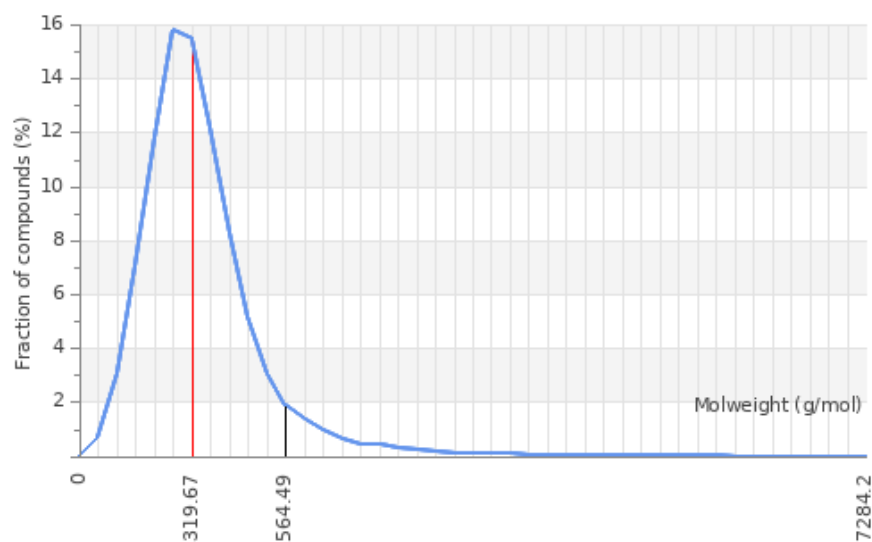
[Print Toxicity Report](#)

Name	OC[C@@H]1O[C@@H]([C@@H]([C@@H]([C@@H]1O)O)O)c1c(O)C([C@H]([C@H]2O)O)O)c2c1oc
Molweight	564.49
Number of hydrogen bond acceptors	13
Number of hydrogen bond donors	10
Number of atoms	40
Number of bonds	44
Number of rotatable bonds	4
Molecular refractivity	133.26
Topological Polar Surface Area	250.97
octanol/water partition coefficient(logP)	-1.75

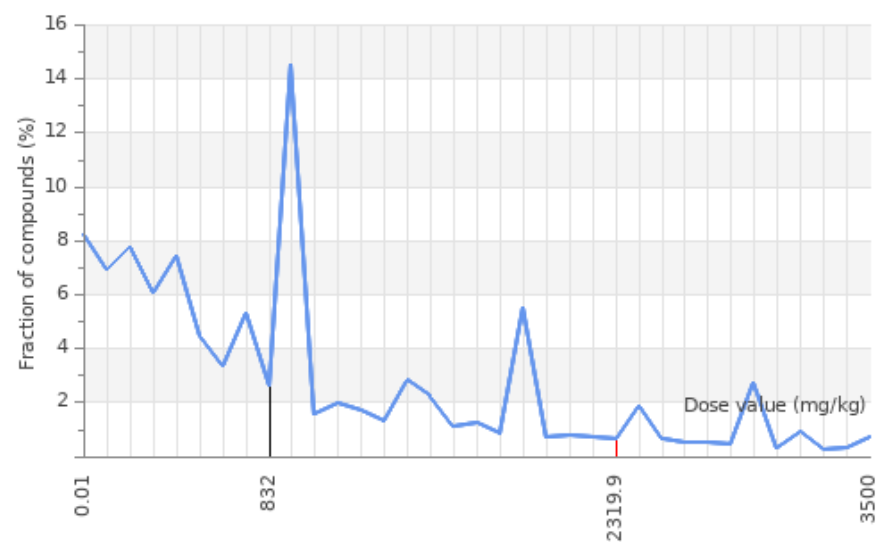
Comparison of input compound with dataset compounds

Value of input compound
 Mean value of dataset

Distribution of molweight



Distribution of dose value



+

Similar compounds

Toxicity Model Report

Copy

Excel

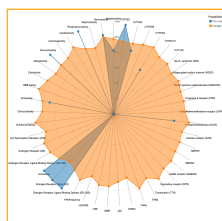
CSV

PDF

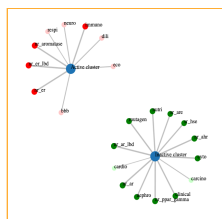
Classification	Target	Shorthand	Prediction	Probability
Organ toxicity	Hepatotoxicity	dili	Inactive	0.86
Organ toxicity	Neurotoxicity	neuro	Inactive	0.88
Organ toxicity	Nephrotoxicity	nephro	Active	0.7
Organ toxicity	Respiratory toxicity	respi	Active	0.73
Organ toxicity	Cardiotoxicity	cardio	Active	0.54
Toxicity end points	Carcinogenicity	carcino	Inactive	0.74

Classification	Target	Shorthand	Prediction	Probability
Toxicity end points	<u>Immunotoxicity</u>	immuno	Active	0.67
Toxicity end points	<u>Mutagenicity</u>	mutagen	Inactive	0.53
Toxicity end points	<u>Cytotoxicity</u>	cyto	Inactive	0.74
Toxicity end points	<u>BBB-barrier</u>	bbb	Inactive	0.53
Toxicity end points	<u>Ecotoxicity</u>	eco	Inactive	0.61
Toxicity end points	<u>Clinical toxicity</u>	clinical	Active	0.56
Toxicity end points	<u>Nutritional toxicity</u>	nutri	Active	0.51
Tox21-Nuclear receptor signalling pathways	<u>Aryl hydrocarbon Receptor (AhR)</u>	nr_ahr	Inactive	0.92
Tox21-Nuclear receptor signalling pathways	<u>Androgen Receptor (AR)</u>	nr_ar	Active	0.56
Tox21-Nuclear receptor signalling pathways	<u>Androgen Receptor Ligand Binding Domain (AR-LBD)</u>	nr_ar_lbd	Inactive	0.99
Tox21-Nuclear receptor signalling pathways	<u>Aromatase</u>	nr_aromatase	Inactive	0.97
Tox21-Nuclear receptor signalling pathways	<u>Estrogen Receptor Alpha (ER)</u>	nr_er	Active	0.50
Tox21-Nuclear receptor signalling pathways	<u>Estrogen Receptor Ligand Binding Domain (ER-LBD)</u>	nr_er_lbd	Inactive	0.99
Tox21-Nuclear receptor signalling pathways	<u>Peroxisome Proliferator Activated Receptor Gamma (PPAR-Gamma)</u>	nr_ppar_gamma	Inactive	0.98
Tox21-Stress response pathways	<u>Nuclear factor (erythroid-derived 2)-like 2/antioxidant responsive element (nrf2/ARE)</u>	sr_are	Inactive	0.94
Tox21-Stress response pathways	<u>Heat shock factor response element (HSE)</u>	sr_hse	Inactive	0.94
Tox21-Stress response pathways	<u>Mitochondrial Membrane Potential (MMP)</u>	sr_mmp	Inactive	0.94
Tox21-Stress response pathways	<u>Phosphoprotein (Tumor Suppressor).p53</u>	sr_p53	Inactive	0.86
Tox21-Stress response pathways	<u>ATPase family AAA domain-containing protein 5 (ATAD5)</u>	sr_atad5	Inactive	0.99
Molecular Initiating Events	<u>Thyroid hormone receptor alpha (THRα)</u>	mie_thr_alpha	Inactive	0.76
Molecular Initiating Events	<u>Thyroid hormone receptor beta (THRβ)</u>	mie_thr_beta	Inactive	0.83
Molecular Initiating Events	<u>Transthyretin (TTR)</u>	mie_ttr	Active	0.78
Molecular Initiating Events	<u>Ryanodine receptor (RYR)</u>	mie_ryr	Inactive	0.83
Molecular Initiating Events	<u>GABA receptor (GABAR)</u>	mie_gabar	Inactive	0.62
Molecular Initiating Events	<u>Glutamate N-methyl-D-aspartate receptor (NMDAR)</u>	mie_nmdar	Inactive	0.97
Molecular Initiating Events	<u>alpha-amino-3-hydroxy-5-methyl-4-isoxazolepropionate receptor (AMPA)</u>	mie_ampar	Inactive	0.99
Molecular Initiating Events	<u>Kainate receptor (KAR)</u>	mie_kar	Inactive	1.0
Molecular Initiating Events	<u>Achetylcholinesterase (AChE)</u>	mie_ache	Inactive	0.51

Classification	Target	Shorthand	Prediction	Probability
Molecular Initiating Events	<u>Constitutive androstane receptor (CAR)</u>	mie_car	Inactive	0.99
Molecular Initiating Events	<u>Pregnane X receptor (PXR)</u>	mie_pxr	Inactive	0.67
Molecular Initiating Events	<u>NADH-quinone oxidoreductase (NADHox)</u>	mie_nadhox	Inactive	0.80
Molecular Initiating Events	<u>Voltage gated sodium channel (VGSC)</u>	mie_vgsc	Inactive	0.89
Molecular Initiating Events	<u>Na⁺/I⁻ symporter (NIS)</u>	mie_nis	Inactive	0.74
Metabolism	<u>Cytochrome CYP1A2</u>	CYP1A2	Inactive	0.86
Metabolism	<u>Cytochrome CYP2C19</u>	CYP2C19	Inactive	0.91
Metabolism	<u>Cytochrome CYP2C9</u>	CYP2C9	Inactive	0.73
Metabolism	<u>Cytochrome CYP2D6</u>	CYP2D6	Inactive	0.91
Metabolism	<u>Cytochrome CYP3A4</u>	CYP3A4	Inactive	0.98
Metabolism	<u>Cytochrome CYP2E1</u>	CYP2E1	Inactive	0.98



The toxicity radar chart is intended to quickly illustrate the confidence of positive toxicity results compared to the average of its class.
Click the thumbnail to access the plot once it has finished loading.



The network chart is intended to quickly illustrate the connection between the selected compound and predicted activities.
Click the thumbnail to access the plot once it has finished loading.

Toxicity targets

Possible binding to toxicity targets is shown below. For more information on the targets, please click on the individual abbreviations.



<u>AA2AR</u>	<u>ADRB2</u>	<u>ANDR</u>	<u>AOFA</u>	<u>CRFR1</u>	<u>DRD3</u>	<u>ESR1</u>	<u>ESR2</u>	<u>GCR</u>	<u>HRH1</u>	<u>NR1I2</u>	<u>OPRK</u>	<u>OPRM</u>	<u>PDE4D</u>	<u>PGH1</u>	<u>PRGR</u>

Last updated: May 2024Disclaimer: Compound structures submitted will not be released under any circumstances. This work is licensed under a [Creative Commons License](#).